Honors Geometry Practice Test 4 — Answer Key

Problems 116–168

Part 1: Essential Skills

Skill 1: Vectors between points

a. $[2-(-4),\ -2-6] = [6,-8]$, with length $\sqrt{36+64} = 10$.

b. $64 + 4 = 68$, so $[8, 2]$ works (also $[2, 8]$, $[-8, 2]$, and so on).

c. No. We'd need integers with $a^2 + b^2 = 21$. Subtracting each square up to 21 leaves $21, 20, 17, 12, 5$, and none of those is a perfect square.

Skill 2: Rationalizing denominators

a. $\dfrac{10}{\sqrt{5}} \cdot \dfrac{\sqrt{5}}{\sqrt{5}} = \dfrac{10\sqrt{5}}{5} = 2\sqrt{5}$

b. $\dfrac{4}{\sqrt{7}-\sqrt{3}} \cdot \dfrac{\sqrt{7}+\sqrt{3}}{\sqrt{7}+\sqrt{3}} = \dfrac{4(\sqrt{7}+\sqrt{3})}{7-3} = \sqrt{7}+\sqrt{3}$

c. $\dfrac{\sqrt{3}}{\sqrt{3}+1} \cdot \dfrac{\sqrt{3}-1}{\sqrt{3}-1} = \dfrac{3-\sqrt{3}}{3-1} = \dfrac{3-\sqrt{3}}{2}$

Skill 3: Scaling a vector

a. $|[5,-12]| = 13$. Length 1: $\left[\frac{5}{13}, -\frac{12}{13}\right]$. Opposite direction, length 26: $-26\left[\frac{5}{13}, -\frac{12}{13}\right] = [-10, 24]$.

b. $|[3,-1]| = \sqrt{10}$. Length 1: $\left[\dfrac{3}{\sqrt{10}}, -\dfrac{1}{\sqrt{10}}\right] = \left[\dfrac{3\sqrt{10}}{10}, -\dfrac{\sqrt{10}}{10}\right]$. Opposite direction, length 5: $-5\left[\dfrac{3\sqrt{10}}{10}, -\dfrac{\sqrt{10}}{10}\right] = \left[-\dfrac{3\sqrt{10}}{2}, \dfrac{\sqrt{10}}{2}\right]$.

Skill 4: Perpendicular vectors

a. $[1,-7]$ and $[-1, 7]$. Each has length $\sqrt{50} = 5\sqrt{2}$ already.

b. $[6, 8]$ and $[-6,-8]$ are perpendicular to $[-8, 6]$ with length 10. Doubling: $[12, 16]$ and $[-12,-16]$.

Skill 5: A point part of the way along a segment

a. $\overrightarrow{AB} = [12,-8]$, so the point is $(-5, 6) + \frac{3}{4}[12,-8] = (-5+9,\ 6-6) = (4, 0)$.

b. $\overrightarrow{AP} = [8,-6] = \frac{2}{3}\overrightarrow{AB}$, so $\overrightarrow{AB} = [12,-9]$ and $B = (8, 0)$.

Skill 6: Adding vectors

a. $[5,-2] + [1, 6] = [6, 4]$, so the return vector is $[-6,-4]$.

b. Components: $2s + t = 7$ and $s + 3t = 11$. Solving gives $s = 2$, $t = 3$. Check: $2[2, 1] + 3[1, 3] = [4 + 3,\ 2 + 9] = [7, 11]$.

Skill 7: Special right triangles

a.

Triangle	Short leg	Long leg	Hypotenuse
30-60-90	9	$9\sqrt{3}$	18
30-60-90	6	$6\sqrt{3}$	12
30-60-90	$4\sqrt{3}$	12	$8\sqrt{3}$
30-60-90	7	$7\sqrt{3}$	14

Triangle	Leg	Leg	Hypotenuse
45-45-90	7	7	$7\sqrt{2}$
45-45-90	$9\sqrt{2}$	$9\sqrt{2}$	18

b. The altitude splits the equilateral triangle into two 30-60-90 triangles. Each has hypotenuse 10 (a side) and short leg 5 (half a side), so the altitude is the long leg: $5\sqrt{3}$.

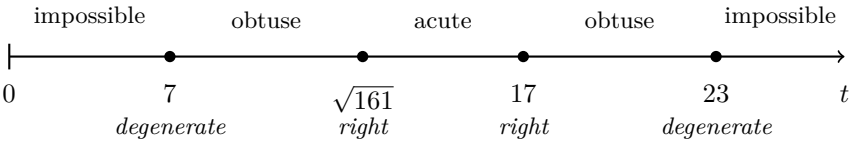
Skill 8: Perimeter of a special right triangle

a. $x + x\sqrt{3} + 2x = 18$, so $x(3 + \sqrt{3}) = 18$ and $x = \dfrac{18}{3 + \sqrt{3}} \cdot \dfrac{3 - \sqrt{3}}{3 - \sqrt{3}} = \dfrac{18(3 - \sqrt{3})}{6} = 9 - 3\sqrt{3}$. Sides: short leg $9 - 3\sqrt{3}$, long leg $9\sqrt{3} - 9$, hypotenuse $18 - 6\sqrt{3}$.

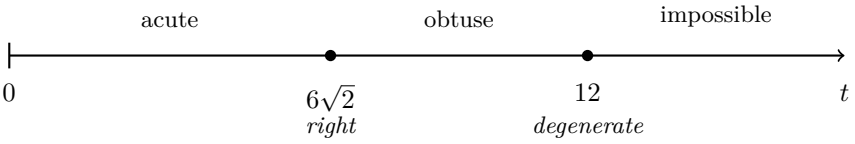
b. $x(2 + \sqrt{2}) = 10$, so $x = \dfrac{10(2 - \sqrt{2})}{2} = 10 - 5\sqrt{2}$. Sides: legs $10 - 5\sqrt{2}$, hypotenuse $10\sqrt{2} - 10$.

Skill 9: Triangle type from side lengths

a. Degenerate: $t + 8 = 15$, so $t = 7$; and $8 + 15 = t$, so $t = 23$. Right with 15 longest: $t^2 + 64 = 225$, so $t = \sqrt{161}$. Right with t longest: $64 + 225 = t^2$, so $t = 17$.



b. Right: $6^2 + 6^2 = t^2$, so $t = 6\sqrt{2}$. Degenerate: $6 + 6 = t$, so $t = 12$.



The first four stages disappear: impossible, degenerate, obtuse, and the first right. The lower degenerate boundary would be $t + 6 = 6$, so $t = 0$, which isn't a triangle. The early obtuse and right stages need a fixed side to be the longest side, opposite the obtuse or right angle; with two equal fixed sides that angle would appear twice, and a triangle can't have two right or obtuse angles.

c. Answers vary. Acute: 5, 6, 7 ($5 + 6 > 7$, and $5^2 + 6^2 = 61 > 49 = 7^2$). Obtuse: 4, 5, 8 ($4 + 5 > 8$, and $4^2 + 5^2 = 41 < 64 = 8^2$). Any set must pass the triangle inequality (the two shorter sides add to more than the longest), then compare $a^2 + b^2$ with c^2 for the longest side c.

Skill 10: Reflections

a. $(x, y) \to (-y, -x)$: $A' = (-2, -1)$, $B' = (-3, -4)$, $C' = (-5, -2)$.

b. Midpoint: $(1, 2)$. The vector $[8, -4]$ has slope $-\frac{1}{2}$, so the mirror line has slope 2: $y - 2 = 2(x - 1)$, or $y = 2x$.

Skill 11: Identifying a transformation

$A \to B \to C$ goes counterclockwise.

i. Translation by $[3, 4]$: every vertex moves by $[3, 4]$, and the order stays counterclockwise.

ii. Reflection over $x = 5$: the order is reversed, and $x = 5$ is the perpendicular bisector of each pair, with midpoints $(5, 1)$, $(5, 1)$, $(5, 3)$ and every segment horizontal.

iii. Rotation: the order is kept (still counterclockwise), but the vertices don't all move by the same vector (A moves by $[-2, 0]$, B by $[-5, 3]$), so it isn't a translation. (It's a $90°$ turn about the origin; students only need to recognize it as a rotation.)

iv. Glide-reflection: the order is reversed, but there's no single mirror line. The perpendicular bisector of AA' is $y = 2x - 6$ (through midpoint $(3, 0)$ with slope 2), but the midpoint of BB', $(6, 0)$, isn't on it. It's a reflection over the x-axis followed by a translation by $[4, 0]$.

Skill 12: Angle hunts

a. $a = 58°$ (vertical to the $58°$ angle at P). $b = 180° - 125° = 55°$ (linear pair at Q). $c = 180° - 58° - 55° = 67°$ (triangle angle sum). $d = 180° - 67° = 113°$ (linear pair at R).

b. $d = 113° = 58° + 55°$: the exterior angle at R equals the sum of the two interior angles at the other vertices. Both d and $58° + 55°$ are what's left when c is subtracted from $180°$ (one by the linear pair, one by the triangle sum), so they must be equal.

Part 2: Putting Skills Together

Question 1

a. $\overrightarrow{AB} = [4, 2]$, so the side length is $\sqrt{20} = 2\sqrt{5}$.

b. The altitude splits the triangle into two 30-60-90 triangles with short leg $\sqrt{5}$ (half of a side), so the altitude is $\sqrt{5} \cdot \sqrt{3} = \sqrt{15}$.

c. The altitude starts at the midpoint $M = (1, 3)$ and is perpendicular to $[4, 2]$, so it points along $[-1, 2]$ or $[1, -2]$, each of length $\sqrt{5}$. Scale to length $\sqrt{15}$: $\frac{\sqrt{15}}{\sqrt{5}}[-1, 2] = \sqrt{3}[-1, 2] = [-\sqrt{3}, 2\sqrt{3}]$. $C = (1 - \sqrt{3},\ 3 + 2\sqrt{3})$ or $C = (1 + \sqrt{3},\ 3 - 2\sqrt{3})$. Check: from A to the first C is $[2 - \sqrt{3},\ 1 + 2\sqrt{3}]$, with length squared $(7 - 4\sqrt{3}) + (13 + 4\sqrt{3}) = 20$.

Question 2

$\overrightarrow{AB} = [3, 6]$, with length $\sqrt{45} = 3\sqrt{5}$. Length 1 in that direction: $\left[\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right]$. Length 5: $\left[\frac{5}{\sqrt{5}}, \frac{10}{\sqrt{5}}\right] = [\sqrt{5}, 2\sqrt{5}]$. The point is $(1 + \sqrt{5},\ 2 + 2\sqrt{5})$.

Question 3

a. Right angle at A: $\overline{AC}$ must be vertical, so $t = 0$. Right angle at B: $\overline{BC}$ must be vertical, so $t = 8$. Right angle at C: $CA^2 + CB^2 = AB^2$: $(t^2 + 16) + ((t-8)^2 + 16) = 64$ $2t^2 - 16t + 96 = 64$ $t^2 - 8t + 16 = 0$ $(t - 4)^2 = 0$, so $t = 4$ only. (Slopes work too: $\frac{4}{t} \cdot \frac{4}{t-8} = -1$ gives the same equation.)

b. Obtuse: $t < 0$ (obtuse at A) or $t > 8$ (obtuse at B). Acute: $0 < t < 4$ or $4 < t < 8$. In this range the angles at A and B are acute, and $CA^2 + CB^2 - AB^2 = 2(t - 4)^2 > 0$, so the angle at C is acute too. The angle at C reaches $90°$ only at $t = 4$.

Question 4

Let D be the foot of the altitude. In $\triangle ABD$ (45-45-90): $BD = 6\sqrt{3}$ and $AB = 6\sqrt{3} \cdot \sqrt{2} = 6\sqrt{6}$. In $\triangle ACD$ (30-60-90): $AD = 6\sqrt{3}$ is the long leg, so $CD = 6$ and $AC = 12$. Sides: $AB = 6\sqrt{6}$, $AC = 12$, $BC = 6\sqrt{3} + 6$. Perimeter: $18 + 6\sqrt{3} + 6\sqrt{6}$.

Question 5

a. Put the corner at the origin with the walls along the axes. The center is (r, r) and $P = (2, 9)$. The right triangle from the center to P has legs $r - 2$ and $r - 9$, and its hypotenuse is a radius, r.

b. $(r - 2)^2 + (r - 9)^2 = r^2$ $r^2 - 4r + 4 + r^2 - 18r + 81 = r^2$ $r^2 - 22r + 85 = 0$ $(r - 5)(r - 17) = 0$, so $r = 5$ or $r = 17$.

c. $r = 17$ inches. If $r = 5$, the center would be 5 inches from each wall, but P is 9 inches from one wall, farther than the center. Then P would be on the side of the table facing away from the corner, not in the gap between the table and the corner.

Question 6

a. $KL = |[1, 7]| = \sqrt{50}$, $LM = |[-3, -1]| = \sqrt{10}$, $MK = |[2, -6]| = \sqrt{40}$. $XZ = |[5, 5]| = \sqrt{50}$, $ZY = |[1, -3]| = \sqrt{10}$, $YX = |[-6, -2]| = \sqrt{40}$. All three pairs of corresponding sides are equal, so the triangles are congruent.

b. $K \to L \to M$ goes counterclockwise, but $X \to Z \to Y$ goes clockwise. Rotations and translations keep the order, so this has to be a reflection (or glide-reflection).

c. Midpoint of KX: $(-2, 1)$. $\overrightarrow{KX} = [4, -2]$ has slope $-\frac{1}{2}$, so the mirror line has slope 2: $y - 1 = 2(x + 2)$, or $y = 2x + 5$. Check: the midpoint of LZ is $(1, 7)$ and $2(1) + 5 = 7$; the midpoint of MY is $(0, 5)$ and $2(0) + 5 = 5$. Also, $\overrightarrow{LZ} = [8, -4]$ and $\overrightarrow{MY} = [12, -6]$ both have slope $-\frac{1}{2}$, perpendicular to the mirror line.

Question 7

a. P: over $x = 1$ gives $(0, 5)$, then over $x = 4$ gives $(8, 5)$. Q: over $x = 1$ gives $(5, 0)$, then over $x = 4$ gives $(3, 0)$. Both moved by $[6, 0]$: the single transformation is a translation by $[6, 0]$, twice the distance between the lines. Opposite order: a translation by $[-6, 0]$.

b. Undo the reflection: $(1, 6) \to (-1, 6)$. Undo the translation: $(-1, 6) + [5, -4] = (4, 2)$. The point started at $(4, 2)$. To undo the sequence: reflect over the y-axis, then translate by $[5, -4]$ (the steps reversed, in reverse order).